



# ■ CS177 Bioinformatics: Software Development and Applications

## Lecture 20: Biological Databases

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Spring, 2024





# Learning objectives



- At the end of this lecture, you will be able to:
  - Explore centralized biological databases: NCBI, EBI
  - Describe main genome browsers and use them to explore genomic data and annotations
  - Search for publications in biomedical literature
  - Search for biological data from diverse sources





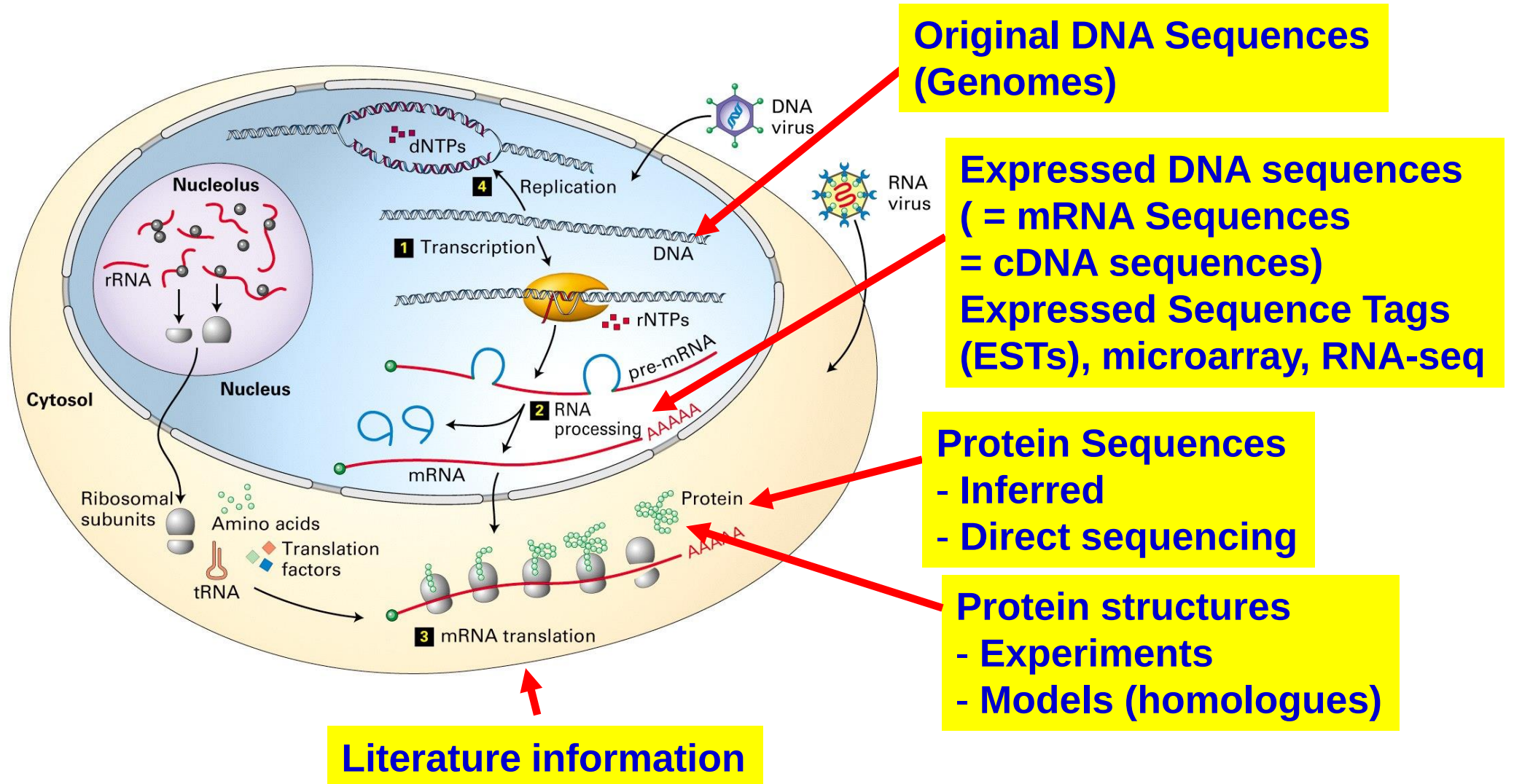
# Introduction to Biological Databases



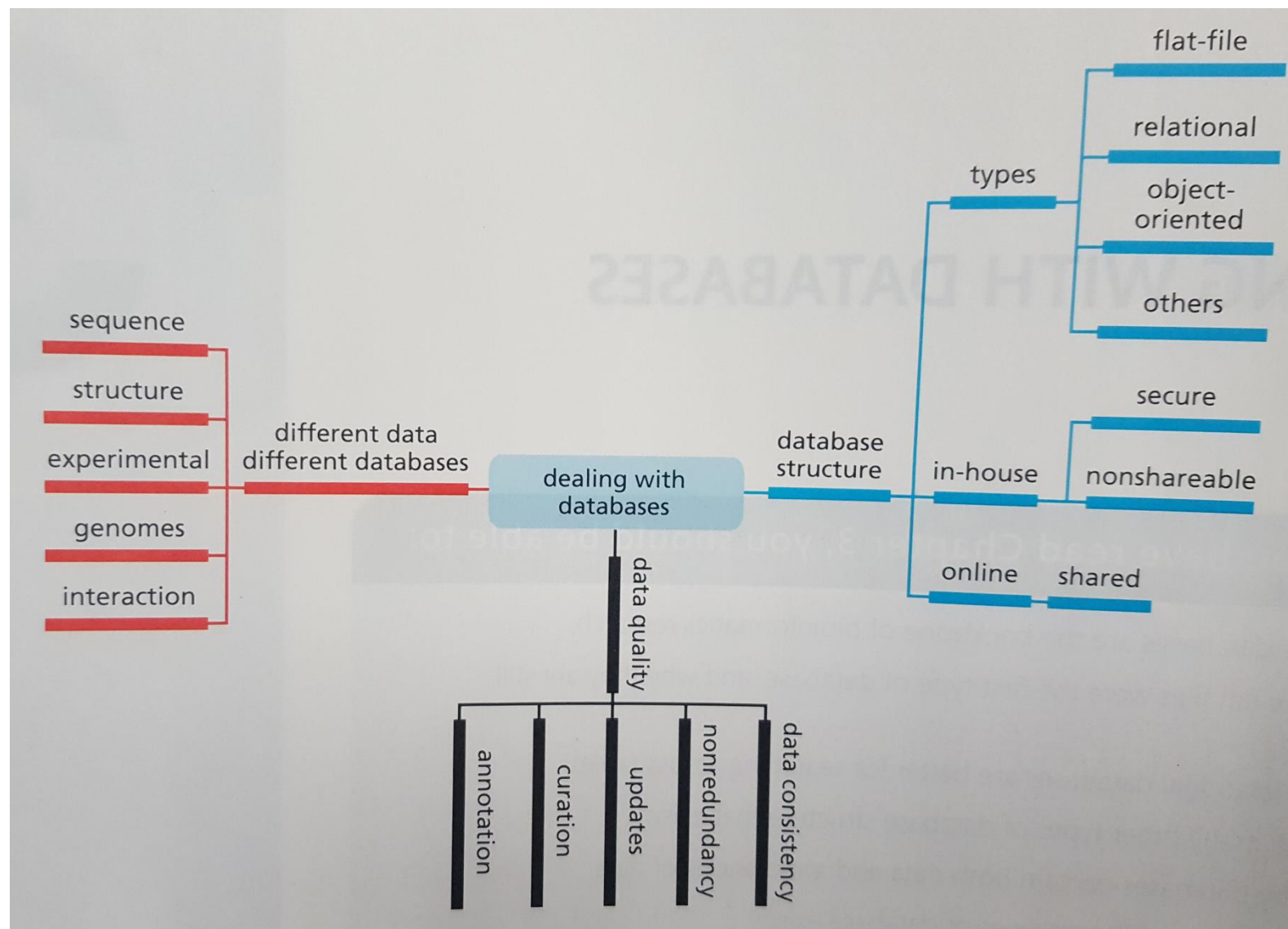
# The Central Dogma & biological data



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# Important & general aspects of databases



From Zvelebil & Baum  
(2008), *Understanding  
Bioinformatics*, page 46



# Biological databases: two perspectives



1. **Individual genes/proteins:** We might want to study only **one** gene, protein, DNA molecule, or other type of object in a database. For example, for human beta globin, there is a gene (*HBB*), a protein sequence, a protein structure, and entries for various kinds of variation
2. **Large-scale queries:** We can think about **large** groups, such as all the globin genes in the human genome, or all the known *HBB* variants. Or we might want to study a set of 100 genes previously implicated in a disease (e.g. autism) to assess their variation in patient samples

These are different ways of thinking about searching databases





# Centralized Databases

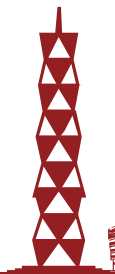
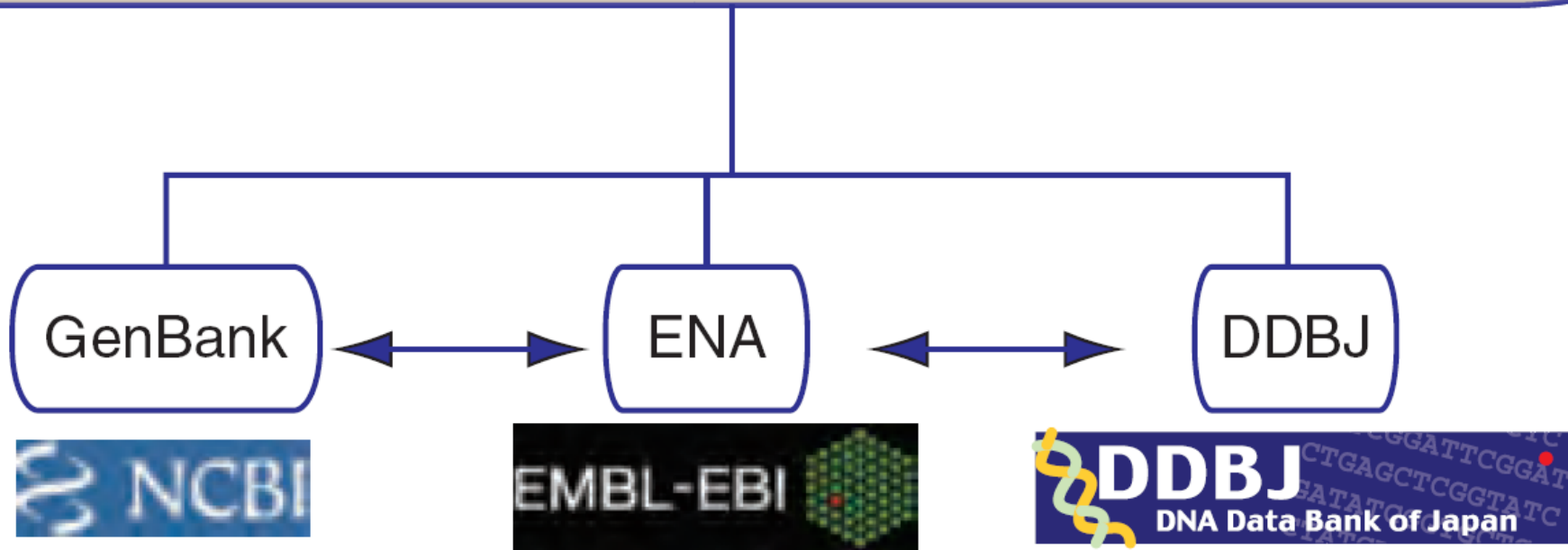




# INSDC coordinates sequence data



International Nucleotide Sequence Database Collaboration (INSDC)





# National Center for Biotechnology Information (NCBI): organization



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*Access via web browser,  
NCBI E-Utils, EDirect*

Entrez system (40 molecular and literature databases)

GenBank

Trace Archive

Sequence Read  
Archive (SRA)

Nucleotide

Genome Survey  
Sequences (GSS)

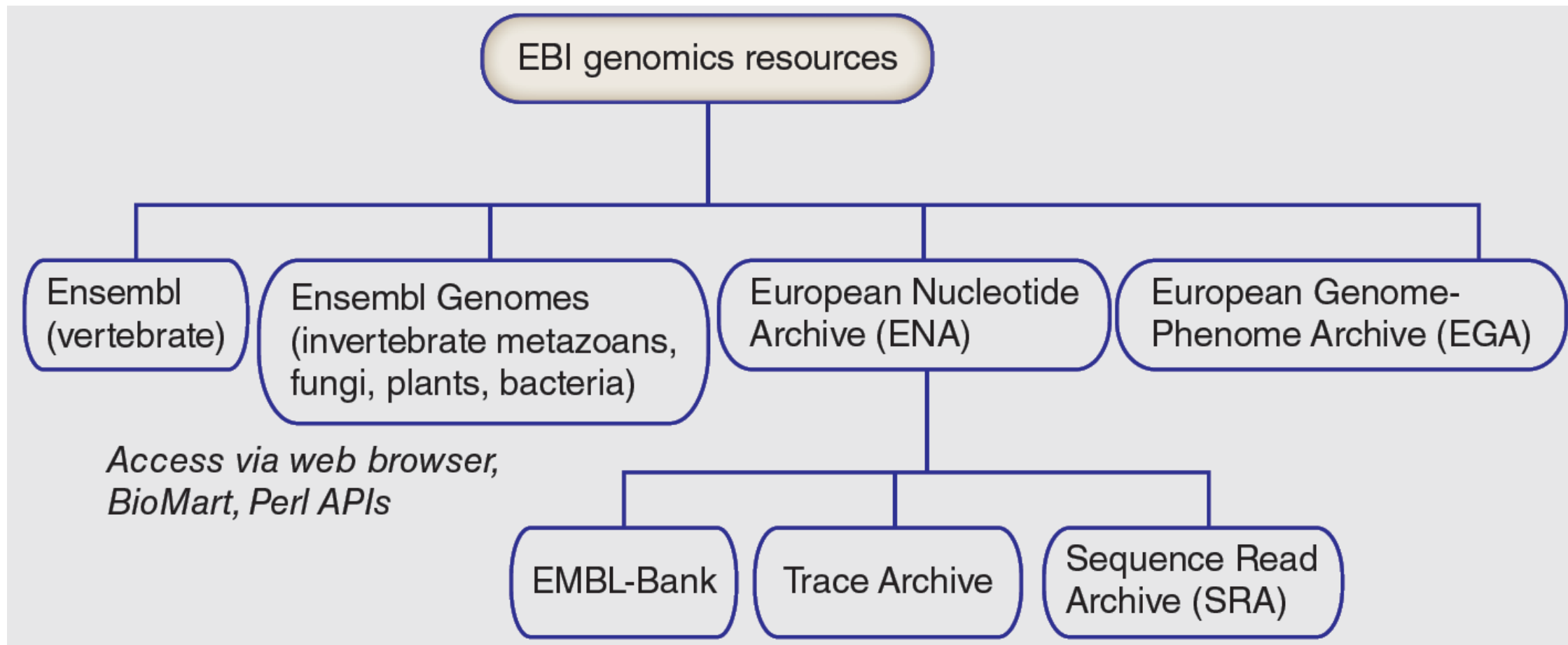
Expressed Sequence  
Tag (EST)



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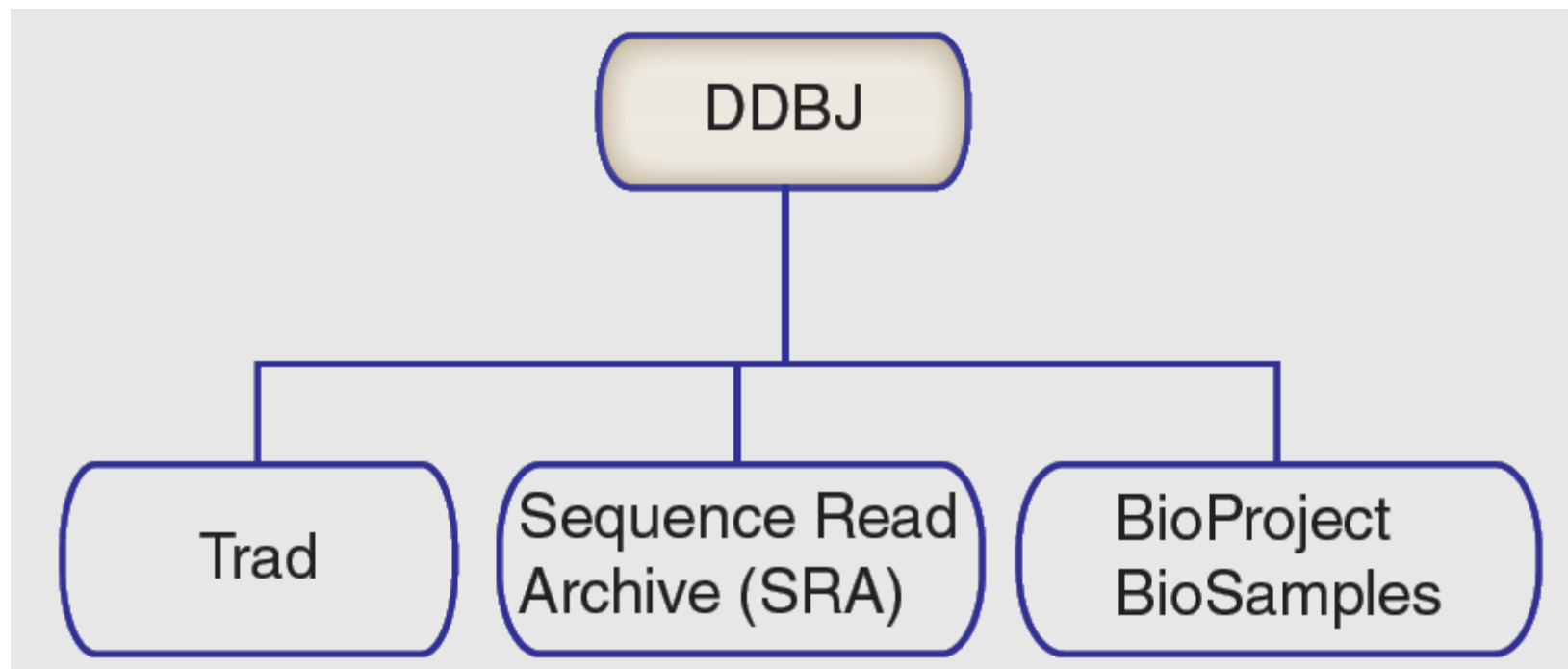


# European Bioinformatics Institute (EBI): organization





# DNA Database of Japan (DDBJ): organization



# Location and history of NCBI



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David J. Lipman, M.D.  
As NCBI Director  
(1989 – 2017)

Created in 1988 as a part of the National Library of Medicine (NLM) at NIH

- Establish public databases
- Research in computational biology
- Develop software tools for sequence analysis
- Disseminate biomedical information



# NCBI Data Model



- [www.ncbi.nlm.nih.gov](http://www.ncbi.nlm.nih.gov)
- In 1990s, the National Center for Biotechnology Information (NCBI) introduced a new model for sequence-related information
  - Centered on a biological sequence as a simple, linear coordinate system
  - Diverse data and information can be closely linked to the coordinate system
- The model resulted in the rapid development of software and the integration of databases that underlie the popular **Entrez** retrieval system
- Advantages:
  - The ability to move effortlessly from the published literature to DNA sequences to the proteins they encode, to chromosome maps of the genes, and to the 3-D structures of the proteins
  - Associating diverse data with the sequence





# Central bioinformatics resource: NCBI



NCBI (with Ensembl, EBI, UCSC) is one of the central bioinformatics websites. It includes:

- PubMed
- Entrez search engine integrating ~40 databases
- BLAST (Basic Local Alignment Search Tool)
- Online Mendelian Inheritance in Man (OMIM)
- Taxonomy
- Books
- Many additional resources





All Databases ▾

Search

NCBI Home

Resource List (A-Z)

All Resources

Chemicals & Bioassays

Data & Software

DNA & RNA

Domains & Structures

Genes & Expression

Genetics & Medicine

Genomes & Maps

Homology

Literature

Proteins

Sequence Analysis

Taxonomy

Training & Tutorials

Variation

## All Resources

All

Databases

Downloads

Submissions

Tools

How To

### Databases

#### [Assembly](#)

A database providing information on the structure of assembled genomes, assembly names and other meta-data, statistical reports, and links to genomic sequence data.

#### [BioCollections](#)

A curated set of metadata for culture collections, museums, herbaria and other natural history collections. The records display collection codes, information about the collections' home institutions, and links to relevant data at NCBI.

#### [BioProject \(formerly Genome Project\)](#)

A collection of genomics, functional genomics, and genetics studies and links to their resulting datasets. This resource describes project scope, material, and objectives and provides a mechanism to retrieve datasets that are often difficult to find due to inconsistent annotation, multiple independent submissions, and the varied nature of diverse data types which are often stored in different databases.

#### [BioSample](#)

The BioSample database contains descriptions of biological source materials used in experimental assays.

#### [Bookshelf](#)

A collection of biomedical books that can be searched directly or from linked data in other NCBI databases. The collection includes biomedical textbooks, other scientific titles, genetic resources such as *GeneReviews*, and NCBI help manuals.

#### [ClinVar](#)

A resource to provide a public, tracked record of reported relationships between human variation and observed health status with supporting evidence. Related information in the [NIH Genetic Testing Registry \(GTR\)](#), [MedGen](#), [Gene](#), [OMIM](#), [PubMed](#) and other sources is accessible through hyperlinks on the records.



# Genome Browsers





# Genome Browsers



- Versatile tools to visualize chromosomal positions (typically on x-axis) with annotation tracks (typically on y-axis)
- Useful to explore data related to some chromosomal feature of interest, such as a gene
- Prominent browsers include UCSC, Ensembl, and NCBI
- Many hundreds of specialized genome browsers are available, some for particular organisms or molecule types

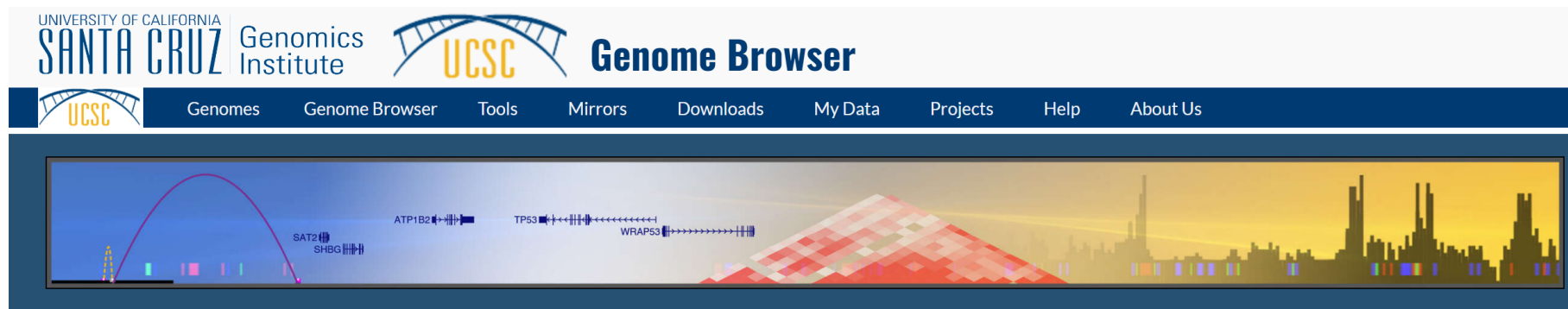


# UCSC Genome Browser



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- <http://genome.ucsc.edu/> (University of California, Santa Cruz)



Search genes, data, help docs and more...

Search

## Tools



hg38



hg19



mm39

- **Genome Browser** - Interactively visualize genomic data
- **BLAT** - Rapidly align sequences to the genome
- **In-Silico PCR** - Rapidly align PCR primer pairs to the genome
- **Table Browser** - Download and filter data from the Genome Browser
- **LiftOver** - Convert genome coordinates between assemblies
- **REST API** - Returns data requested in JSON format
- **Variant Annotation Integrator** - Annotate genomic variants
- **More tools...**

## News

- Apr. 25, 2024 - **New AbSplice Prediction Scores track for hg19**
- Mar. 26, 2024 - **New gnomAD v4 Constraint Metrics (hg38) and gnomAD Non-can**
- Mar. 07, 2024 - **New Prediction Scores super track and BayesDel track for hg19**
- Mar. 05, 2024 - **New JASPAR tracks: Human (hg19/hg38) - Mouse (mm10/mm39)**
- Mar. 01, 2024 - **AbSplice Prediction Scores for hg38**
- Feb. 21, 2024 - **New DECIPHER Dosage Sensitivity tracks for Human (hg19/hg38)**

More news...

Subscribe

## Meetings and Workshops: Come see us in person!

■ **Medical Genetics Center**  
**Netherlands** - Leiden,  
Netherlands - Apr 29, 2024

■ **Eramus Medical Center** -  
Rotterdam, Netherlands - Apr  
30, 2024

■ **European Human Genetics**  
**Conference** - Berlin, Germany  
- Jun 1-4, 2024

■ **Children's Mercy Genomic**  
**Medicine Short Course** -  
Kansas City, MO - Jun 10-13,  
2024

Feel free to [contact us](#) if you are interested in attending a workshop, or meeting someone from the team to collaborate, get help, or ask any questions at the meetings.

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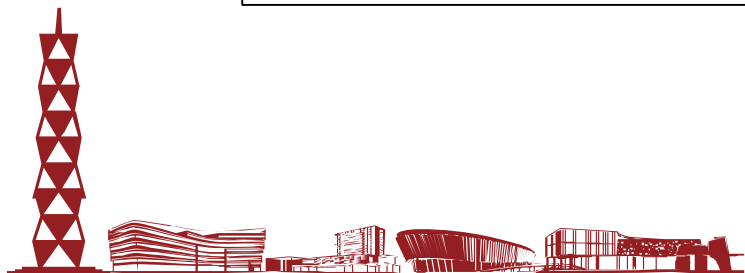
# UCSC Genome Browser



- Choose the group (e.g. mammal), genome (e.g. human), assembly (e.g. GRCh37 or GRCh38), position and/or search term (e.g. hbb)

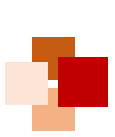
| group                                                                      | genome  | assembly                  | position                    | search term                                                |        |
|----------------------------------------------------------------------------|---------|---------------------------|-----------------------------|------------------------------------------------------------|--------|
| Mammal ▼                                                                   | Human ▼ | Feb. 2009 (GRCh37/hg19) ▼ | chr21:33,031,597-33,041,570 | hbb                                                        | submit |
|                                                                            |         |                           |                             | HBB (Homo sapiens hemoglobin, beta (HBB), mRNA.)           |        |
|                                                                            |         |                           |                             | HBBP1 (Homo sapiens hemoglobin, beta pseudogene 1 (HBBP1), |        |
| <a href="#">Click here to reset</a> the browser user interface settings to |         |                           |                             |                                                            |        |
| track search                                                               |         | add custom tracks         |                             | track hubs                                                 |        |
| configure tracks and display                                               |         |                           |                             |                                                            |        |

A **genome build** or **assembly** (e.g. GRCh37 or GRCh38) refers to a fixed, agreed-upon version of a reference genome. Assemblies are typically updated every few years





Explore the browser! Begin with a favorite gene or region. Zoom in to base pair level, then out to full chromosome level. Explore the many tracks you can add.



# Ensembl genome database and browser



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[www.ensembl.org](http://www.ensembl.org) (EBI and Wellcome Trust Sanger Institute)

 [BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#) | [Blog](#)

Login/Register

 Search all species... 

Tools  
[All tools](#)

BioMart >  
Export custom datasets from Ensembl with this data-mining tool

BLAST/BLAT >  
Search our genomes for your DNA or protein sequence

Variant Effect Predictor >  
Analyse your own variants and predict the functional consequences of known and unknown variants

Search

All species ▼

for

Go

e.g. [BRCA2](#) or [rat 5:62797383-63627669](#) or [rs699](#) or [coronary heart disease](#)

All genomes  
-- Select a species -- ▼



Pig breeds  
Pig reference genome and 12 additional breeds

[View full list of all species](#)

Favourite genomes 



Human  
GRCh38.p14

[Still using GRCh37?](#)



Mouse  
GRCm39



Zebrafish  
GRCz11

Ensembl is a genome browser for vertebrate genomes that supports research in comparative genomics, evolution, sequence variation and transcriptional regulation. Ensembl annotate genes, computes multiple alignments, predicts regulatory function and collects disease data. Ensembl tools include BLAST, BLAT, BioMart and the Variant Effect Predictor (VEP) for all supported species.

## Ensembl Release 111 (January 2024)

- MANE Select (v1.2) GRCh38.p14 patch annotation
- Human variation data updated to dbSNP156
- Updated genome assembly and annotation for sheep and cattle
- Regulatory annotation of open chromatin regions and promoters in common carp and rainbow trout (a collaboration with the AQUA-FAANG consortium)
- Updated regulatory annotation, including enhancers, for pig and chicken, Atlantic salmon, European seabass and turbot

[More release news](#) on our blog

## Ensembl Rapid Release

**New assemblies with gene and protein annotation every two weeks.**

Note: species that already exist on this site will continue to be updated with the full range of annotations.

Go

The Ensembl Rapid Release website provides annotation for recently produced, publicly available vertebrate and non-vertebrate genomes from biodiversity initiatives such as Darwin Tree of Life, the Vertebrate Genomes Project and the Earth BioGenome Project.

[Rapid Release news](#) on our blog



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# Biomedical Literature



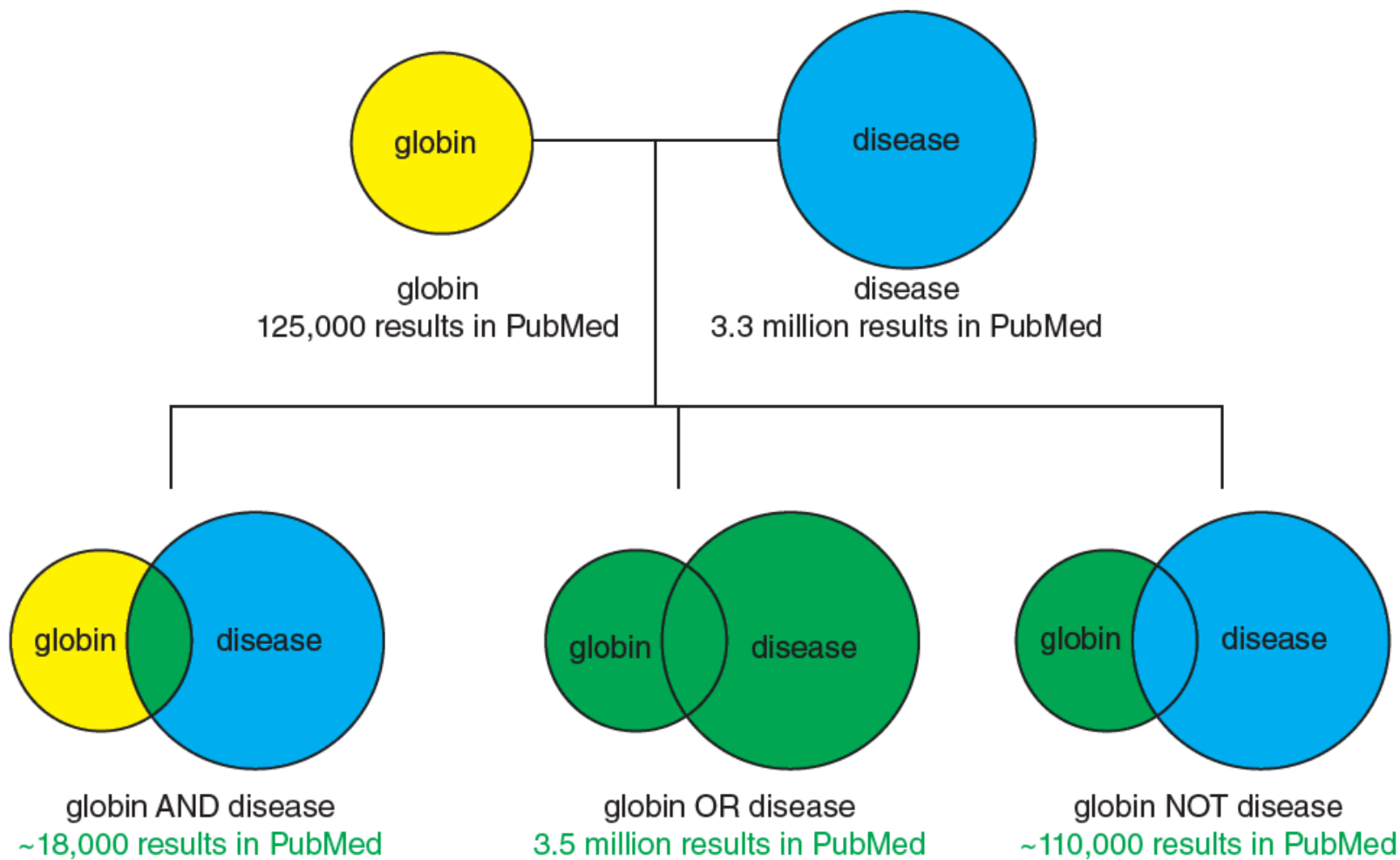


- <https://www.ncbi.nlm.nih.gov/pubmed>
- **PubMed** is a free search engine accessing primarily the **MEDLINE** database: contains bibliographic citations and author abstracts from over 4,600 journals published in the United States and in 70 foreign countries. As of 11 July 2017, it has 27.3 million records dating back to 1966
- **MeSH (Medical Subject Headings)** is the list of the vocabulary terms used for subject analysis of biomedical literature at NLM
  - MeSH vocabulary is used for indexing journal articles for MEDLINE
  - MeSH controlled vocabulary imposes uniformity and consistency to the indexing of biomedical literature
- **PubMed ID**: Identifies articles managed by NCBI
  - Reliable, stable link to citation
  - Example: PMID: 12205585





# Boolean operators for PubMed search



# Diverse sources of data



- Nucleic Acids Research (NAR) Database Issue
- Journals dedicated to data: Database, Scientific Data
- Journal papers' supplementary (online) materials
- Data resources dedicated to specific model organisms, diseases or topics:
  - FlyBase, WormBase, PlasmoDB
  - TCGA (The Cancer Genome Atlas)
  - Cancer Dependency Map (DepMap)
  - AnAge: Animal Ageing and Longevity Database
- International projects:
  - ENCODE, HapMap, 1000 Genomes Project
  - NIH Roadmap Epigenomics
  - Human Cell Atlas
- Notable databases and knowledge bases:
  - PDB (Protein Data Bank), DrugBank
  - Gene Ontology, KEGG, DAVID
  - STRING, BioGRID
  - BioModels

## Nucleic Acids Research

VOLUME 46 DATABASE ISSUE JANUARY 4, 2018  
<https://academic.oup.com/nar>





# Hands-on Exploration of Famous Databases (PDB and DepMap)



# Protein Data Bank in the USA (RCSB PDB)



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RCSB PDB Deposit Search Visualize Analyze Download Learn About Documentation Careers COVID-19 MyPDB Contact us

**RCSB PDB** PROTEIN DATA BANK

219,515 Structures from the PDB  
1,068,577 Computed Structure Models (CSM)

3D Structures ? Enter search term(s), Entry ID(s), or sequence Include CSM ?

Advanced Search | Browse Annotations Help

PDB-101 wwPDB EMDataResource NAKB wwPDB Foundation PDB-Dev

Access Computed Structure Models (CSMs) of available model organisms [Learn more](#)

Welcome

Deposit Search Visualize Analyze Download Learn

RCSB Protein Data Bank (RCSB PDB) enables breakthroughs in science and education by providing access and tools for exploration, visualization, and analysis of:

- Experimentally-determined 3D structures from the **Protein Data Bank (PDB)** archive
- Computed Structure Models (CSM)** from AlphaFold DB and ModelArchive

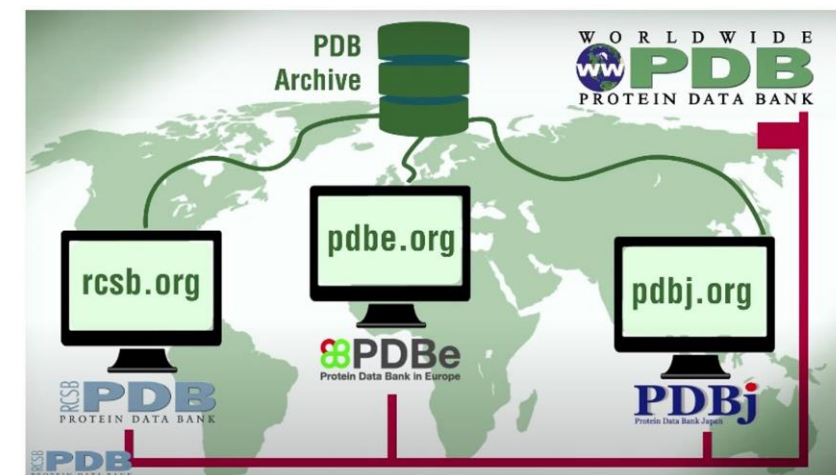
These data can be explored in context of external annotations providing a structural view of biology.

Explore NEW Features

PDB-101 Training Resources

May Molecule of the Month

CFTR and Cystic Fibrosis



Latest Entries As of Tue May 07 2024

Features & Highlights

Register Now for a Webinar on Understanding PDB Validation: Which experimental structures should I rely on?

Join us on May 14 to learn about primary PDB structure quality metrics, including those in the validation slider

News Publications

Poster Prize Awarded at DiscoverBMB

The wwPDB Foundation awarded Angela Kayll (James Madison University) for her poster *Solution Structure of the hMDH2-hCS Metabolon*.

RCSB (Research Collaboratory for Structural Bioinformatics)

<https://www.rcsb.org/>

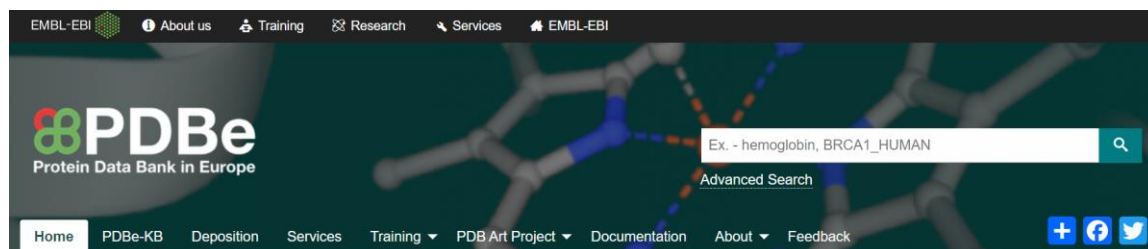
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# Protein Data Bank in Europe (PDBe)



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PDBe is a founding member of the Worldwide Protein Data Bank (wwPDB) which collects, organises and disseminates data on biological macromolecular structures. wwPDB Partners: [RCSB PDB](#), [PDBj](#), [BMRB](#), [EMDB](#). [Read more about PDBe](#).

## Services



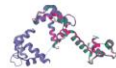
PDBe API



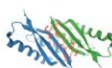
Advanced Search



SIFTS



PDBeFold



PDBePISA



PDBeChem



Download service



Mol\*

[More services](#)

## Latest archive statistics

Total PDB entries in the archive: 219515

### Latest Data:



376

Latest Entries



81

Latest Chemistry



93

Latest Biology

## Featured



## Latest news



Coming soon: Annotation of Protein Modifications in the PDB

2024-05-02

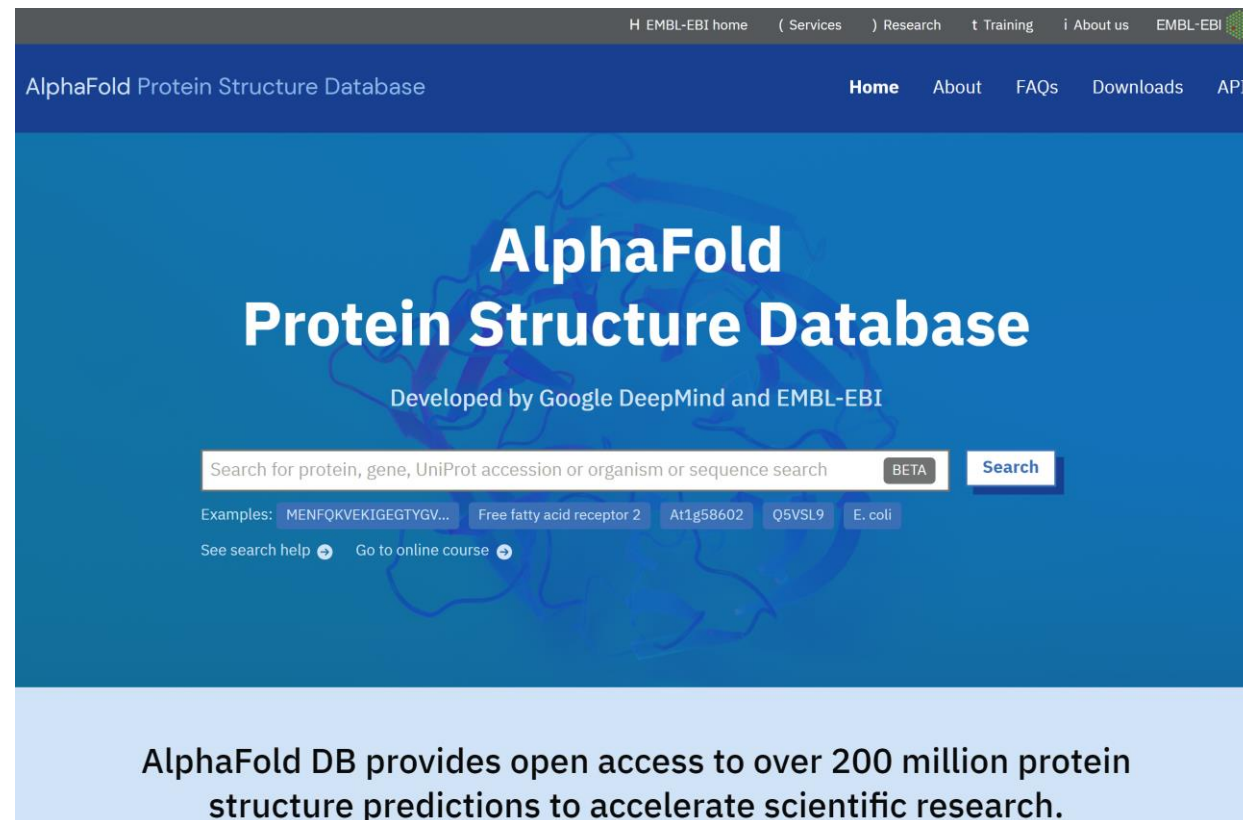
The standardization of protein modification handling ensures that there is a single correct...



CASP16 Call for Targets

2024-04-09

CASP (Critical Assessment of protein...)



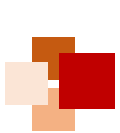
<https://alphafold.ebi.ac.uk/>

<https://alphafold.com/>

<https://www.ebi.ac.uk/pdbe/>



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# Protein Data Bank Japan (PDBj)



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219515

PDB entries from 2024-05-08

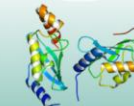


English 日本語 简体中文 繁體中文 한국어

Search pdbj.org

RCSB PDB PDBe BMRB Adv. Search Search help

9BIV



Latest  
new  
entries

## Home

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## Data deposition (OneDep)

Help  
Deposition to PDB, EMDB or BMRB

## Download

Download PDB archive / snapshot  
archive

## Standard format

PDBx/mmCIF Resources  
Format Conversion  
PDBx/mmCIF editor

## Quick links

Help  
non-PDB format compatible  
Group Depositions  
Chemical Component entries  
Latest entries

## About PDBj

**PDBj (Protein Data Bank Japan)** maintains the single global PDB/BMRB/EMDB archives of macromolecular structures and provide integrated tools, under the Joint Usage and Research activities of the Institute for Protein Research. The database is supported by [JST-NBDC](#) (JPMJND2205) and [AMED-BINDS](#). [PDBj's logo](#) design incorporates cytochrome c (PDBID: [1cyc](#)), which was the first structure determined in Japan.

## Suspension of annotation processing during holidays

From Friday, May 3, 2024 to Monday, May 6, 2024 (JST) during the Golden Week holidays, **PDBj will suspend annotation processing of PDB/EMDB/BMRB entries.** [OneDep deposition services will continue to be available.](#)

## Find the service you need

Choose a keyword listed below or input keywords into the textbox at the right of the keyword list. The brief explanation of the matched services will be displayed.

- Click the 'Show all services' button to display the explanation for all services.
- Input some keywords into the 'Word Search Box' to narrow down the search results.

- |                                       |                                              |                                              |                                               |
|---------------------------------------|----------------------------------------------|----------------------------------------------|-----------------------------------------------|
| <input type="checkbox"/> PDB          | <input type="checkbox"/> BMRB                | <input type="checkbox"/> EMD                 |                                               |
| <input type="checkbox"/> search       | <input type="checkbox"/> deposition          | <input type="checkbox"/> viewer              | <input type="checkbox"/> education/dictionary |
| <input type="checkbox"/> NMR          | <input type="checkbox"/> electron microscopy | <input type="checkbox"/> secondary structure | <input type="checkbox"/> sequence             |
| <input type="checkbox"/> similarity   | <input type="checkbox"/> function prediction | <input type="checkbox"/> chemical component  | <input type="checkbox"/> structure prediction |
| <input type="checkbox"/> binding site | <input type="checkbox"/> surface structure   | <input type="checkbox"/> 3D structure        | <input type="checkbox"/> genome               |
| <input type="checkbox"/> RDF          | <input type="checkbox"/> SPARQL              | <input type="checkbox"/> gene                | <input type="checkbox"/> disease              |
| <input type="checkbox"/> drug         |                                              |                                              |                                               |

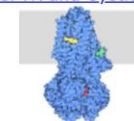
Show all services

e.g. motif, molecular surf

Reset

## Molecule of the Month

[293: CFTR and Cystic Fibrosis](#)



[Article List](#)



## Latest news

- 2024-05-08 [376 new PDB entries have been released on 2024-05-08.](#)
- 2024-04-09 [\[wwPDB\] CASP16 Call for Targets](#)

<https://pdbj.org/>

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**欧洲  
PDBe**

**中国**  
**PDBc**

美国  
RCSB PDB

日本  
PDBj



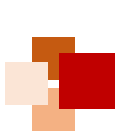


# Hands-on exploration of PDB



- Watch online tutorials, e.g.:
  - [https://www.bilibili.com/video/BV15v4y1M7QJ/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV15v4y1M7QJ/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
  - [https://www.bilibili.com/video/BV1eH4y1M74T/?spm\\_id\\_from=333.788.recommend\\_more\\_video.0&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1eH4y1M74T/?spm_id_from=333.788.recommend_more_video.0&vd_source=4d96844e7fc7e94828f597310890ebb4)
  - There are also tutorials about using PDB on youtube
- Explore the websites of PDB
  - Search and display proteins interesting to you, e.g. those seen in your course projects
- Questions:
  - How can PDB contribute to cancer research and biomedicine?
  - What was the role of PDB in the development of AlphaFold?
  - How to use AI to help build PDB China successfully?





# Cancer Dependency Map (DepMap)

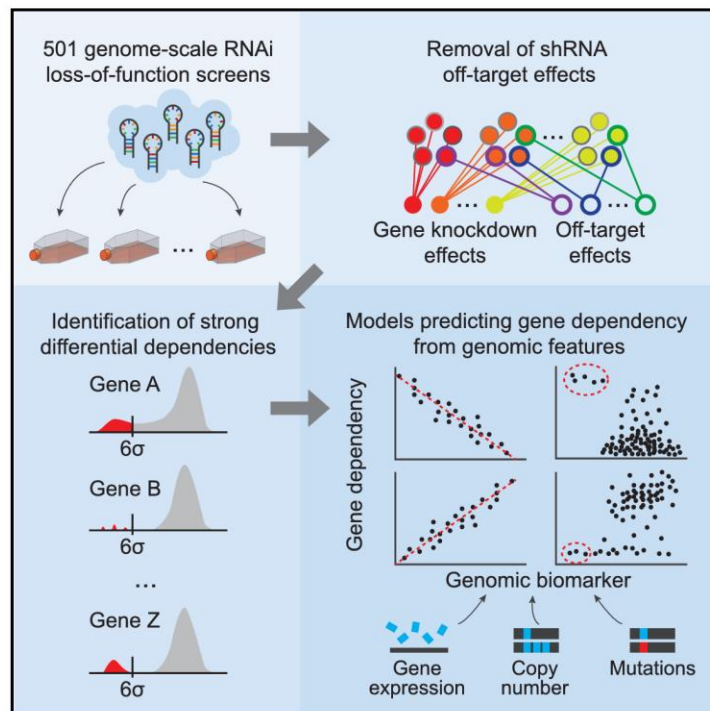


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## Cell

### Defining a Cancer Dependency Map

#### Graphical Abstract



#### Authors

Aviad Tsherniak, Francisca Vazquez, Phil G. Montgomery, ..., Todd R. Golub, Jesse S. Boehm, William C. Hahn

#### Correspondence

william\_hahn@dfci.harvard.edu

#### In Brief

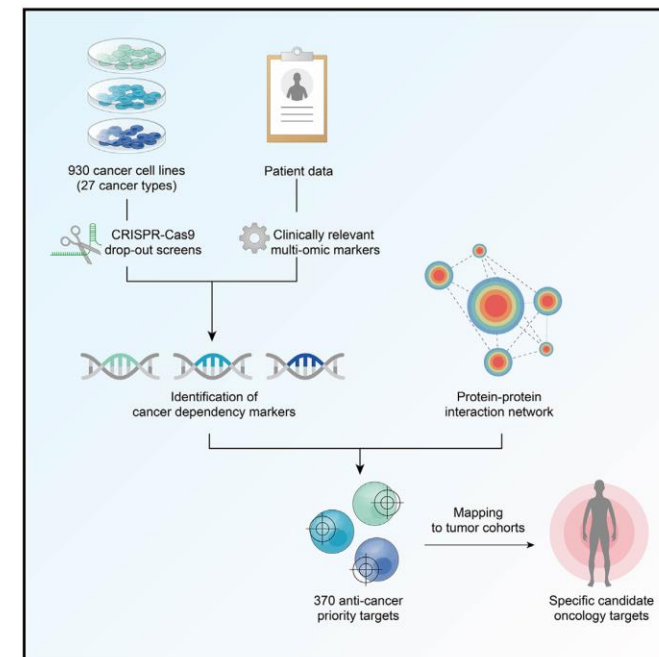
A large-scale analysis of 501 cancer cell lines reveals new vulnerabilities that will help prioritize therapeutic targets

Resource

## Cancer Cell

### A comprehensive clinically informed map of dependencies in cancer cells and framework for target prioritization

#### Graphical abstract



#### Authors

Clare Pacini, Emma Duncan, Emanuel Gonçalves, ..., Leopold Parts, Francesco Iorio, Mathew J. Garnett

#### Correspondence

francesco.iorio@fht.org (F.I.),  
mg12@sanger.ac.uk (M.J.G.)

#### In brief

Pacini et al. identify selective gene dependencies from CRISPR screens in 930 cell lines. Using clinically curated multi-omics data and protein-protein interaction networks, they curate 370 candidate targets and estimate their prevalence in patient tumor cohorts. This enhances understanding of gene dependencies and comprehensively maps targetable cancer dependencies.

Article

Cell 2017

[https://www.cell.com/cell/fulltext/S0092-8674\(17\)30651-7](https://www.cell.com/cell/fulltext/S0092-8674(17)30651-7)

Cancer Cell 2024

[https://www.cell.com/cancer-cell/fulltext/S1535-6108\(23\)00444-0](https://www.cell.com/cancer-cell/fulltext/S1535-6108(23)00444-0)

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# Cancer Dependency Map (DepMap)



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 depmap

ApproachDataConsortiumPublicationsTeam

## Towards mapping the landscape of cancer vulnerabilities across all tumors

Explore the DepMap public data →

[DMC Access](#)

To accelerate precision cancer medicine, we construct systematic key datasets, analytical and visualization tools, using broad panels of cancer models that represent the diversity of human cancers.

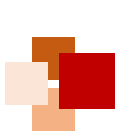


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<https://depmap.org/portal/home/#/>



立志成才报<sup>33</sup>国裕民



# Hands-on exploration of DepMap



- Watch online tutorials, e.g.
  - [https://www.bilibili.com/video/BV1JS4y1d7gX/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1JS4y1d7gX/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
  - Bilibili 公众号 Bioinforbricker 生信进阶之路 | 资源篇 DepMap 系列 (有讲合成致死)
    - [https://www.bilibili.com/video/BV1bL411D7hB/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1bL411D7hB/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
    - [https://www.bilibili.com/video/BV1UM4y127fV/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1UM4y127fV/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
    - [https://www.bilibili.com/video/BV1cM411K7vv/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1cM411K7vv/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
    - [https://www.bilibili.com/video/BV1im4y1y7Fu/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1im4y1y7Fu/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
    - [https://www.bilibili.com/video/BV1xP411r7my/?spm\\_id\\_from=333.337.search-card.all.click&vd\\_source=4d96844e7fc7e94828f597310890ebb4](https://www.bilibili.com/video/BV1xP411r7my/?spm_id_from=333.337.search-card.all.click&vd_source=4d96844e7fc7e94828f597310890ebb4)
- Pick a super-star synthetic lethality (SL), e.g.
  - BRCA-PARP, ATR-ATM, WEE1-TP53, RAD51-AID, PRMT5-MTAP, HRD-USP1, and HRD-Polθ

explore DepMap to find any evidence and biological mechanism of the SL (i.e. to do a case study)





# Recap



- In this lecture, we have learned:
  - Lots of data and resources can be found on-line in several centralized databases, e.g. NCBI, EBI
  - Prominent genome browsers: UCSC, Ensembl, NCBI
  - How to do literature search using PubMed
  - More data can be found on the web: journals, international big projects, etc.
  - Hands-on exploration of PDB and DepMap





# References



- Jonathan Pevsner (2015), *Bioinformatics and Functional Genomics* (3<sup>rd</sup> Edition), Chapter 2 “Access to Sequence Data and Related Information”
- Marketa Zvelebil & Jeremy O. Baum (2008), Understanding Bioinformatics, Chapter 3 “Dealing with Databases”





# Appendix A: Access to GenBank



# ■ How to start a search?



- To acquire information from databases, you may begin with a **query** such as:
  - the name of a protein of interest, or
  - the raw nucleotides comprising a DNA sequence of interest
- DNA sequences and other molecular data are tagged with **accession numbers** that are used to identify a sequence or other record relevant to molecular data





# What is an accession number?



- An **accession number** is a label used to identify a sequence. It is a string of letters and/or numbers that corresponds to a molecular sequence
- Examples:

|              |                                        |
|--------------|----------------------------------------|
| CH471100.2   | GenBank genomic DNA sequence           |
| NC_000001.10 | Genomic contig                         |
| rs121434231  | dbSNP (single nucleotide polymorphism) |

**DNA**

|              |                                         |
|--------------|-----------------------------------------|
| AI687828.1   | An expressed sequence tag (1 of 184)    |
| NM_001206696 | RefSeq DNA sequence (from a transcript) |

**RNA**

|             |                                    |
|-------------|------------------------------------|
| NP_006138.1 | RefSeq protein                     |
| CAA18545.1  | GenBank protein                    |
| O14896      | SwissProt protein                  |
| 1KT7        | Protein Data Bank structure record |

**protein**





# A traditional GenBank record

## The Flat-file Format

```
LOCUS       AY182241                1931 bp    mRNA    linear   PLN 04-MAY-2004
DEFINITION   Malus x domestica (E,E)-alpha-farnesene synthase (AFS1) mRNA,
              complete cds.
ACCESSION    AY182241
VERSION      AY182241.2  GI:32265057
KEYWORDS     .
SOURCE       Malus x domestica (cultivated apple)
  ORGANISM   Malus x domestica
              Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;
              Spermatophyta; Magnoliophyta; eudicotyledons; core eudicots;
              rosids; eurosids I; Rosales; Rosaceae; Maloideae; Malus.
REFERENCE    1 (bases 1 to 1931)
AUTHORS      Pechous,S.W. and Whitaker,B.D.
TITLE        Cloning and functional expression of an (E,E)-alpha-farnesene
              synthase cDNA from peel tissue of apple fruit
JOURNAL      Planta 219, 84-94 (2004)
REFERENCE    2 (bases 1 to 1931)
AUTHORS      Pechous,S.W. and Whitaker,B.D.
TITLE        Direct Submission
JOURNAL      Submitted (18-NOV-2002) PSI-Produce Quality and Safety Lab,
              USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD
              20705, USA
REFERENCE    3 (bases 1 to 1931)
AUTHORS      Pechous,S.W. and Whitaker,B.D.
TITLE        Direct Submission
JOURNAL      Submitted (25-JUN-2003) PSI-Produce Quality and Safety Lab,
              USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD
              20705, USA
REMARK       Sequence update by submitter
COMMENT      On Jun 26, 2003 this sequence version replaced gi:27804758.
FEATURES             Location/Qualifiers
     source          1..1931
                     /organism="Malus x domestica"
                     /mol_type="mRNA"
                     /cultivar="'Law Rome'"
                     /db_xref="taxon:3750"
                     /tissue_type="peel"
     gene            1..1931
                     /gene="AFS1"
     CDS             54..1784
                     /gene="AFS1"
                     /note="terpene synthase"
                     /codon_start=1
                     /product="(E,E)-alpha-farnesene synthase"
                     /protein_id="AAO22848.2"
                     /db_xref="GI:32265058"
                     /translation="MEFRVHLQADNEQKIFQNQMKPEPEASYLINQRRSANYKPNIWK
NDFLDQSLISKYDGEYRKLSEKLEIEVKIYISAETMDLVAKLELIDSVRKLGLANLF
EKEIKEALDSIAAIESDNLGTRDDLYGTALHFKILRQHGKVSQDIFGRFMDKEGTLE
DFLHKNEDLLYNISLIVRLNNDLGTSAAEQERGDSPSSIVCYMREVNASEETARKNIK
GMIDNAWKVNGKCFTTNQVFFLSFMNATNMARVAHSLYKDGDFGDEKGPRTHI
LSLLFQPLVN"
ORIGIN
1  ttcttgtatc  ccaaacatct  cgagcttctt  gtacacccaa  ttaggtatcc  actatggaat
61  tcagagttca  cttgcaagct  gataatgagc  agaaaatttt  tcaaaaccag  atgaaaccgc
121  aacctgaagc  ctcttacttg  attaatacaa  gacggtctgc  aaattacaag  ccaaatatct
181  ggaagaacga  ttctctagat  caatctotta  tcagcaaaata  cgatggagat  gagtatcgga
241  agctgtctga  gaagttaata  gaagaagtta  agatttatat  atctgctgaa  acaatggatt
//
```

Header

Feature Table

Sequence



# The Header



LOCUS AY182241 1931 bp mRNA linear PLN 04-MAY-2004  
DEFINITION Malus x domestica (E,E)-alpha-farnesene synthase (AFS1) mRNA,  
complete cds.  
ACCESSION AY182241  
VERSION AY182241.2 GI:32265057  
KEYWORDS .  
SOURCE Malus x domestica (cultivated apple)  
ORGANISM Malus x domestica  
Eukaryota; Viridiplantae; Streptophyta; Embryophyta; Tracheophyta;  
Spermatophyta; Magnoliophyta; eudicotyledons; core eudicots;  
rosids; eurosids I; Rosales; Rosaceae; Maloideae; Malus.  
REFERENCE 1 (bases 1 to 1931)  
AUTHORS Pechous,S.W. and Whitaker,B.D.  
TITLE Cloning and functional expression of an (E,E)-alpha-farnesene  
synthase cDNA from peel tissue of apple fruit  
JOURNAL Planta 219, 84-94 (2004)  
REFERENCE 2 (bases 1 to 1931)  
AUTHORS Pechous,S.W. and Whitaker,B.D.  
TITLE Direct Submission  
JOURNAL Submitted (18-NOV-2002) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REFERENCE 3 (bases 1 to 1931)  
AUTHORS Pechous,S.W. and Whitaker,B.D.  
TITLE Direct Submission  
JOURNAL Submitted (25-JUN-2003) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REMARK Sequence update by submitter  
COMMENT On Jun 26, 2003 this sequence version replaced gi:27804758.

# Header: Locus Line



| LOCUS AY182241 1931 bp mRNA linear PLN 04-MAY-2004                                                                                                                                 |                                                                                                                                          |                |      |          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|----------|
| complete cds.                                                                                                                                                                      |                                                                                                                                          |                |      |          |
| ACCESSION                                                                                                                                                                          | AY182241                                                                                                                                 | Length 1931 bp | mRNA | linear   |
| VERSION                                                                                                                                                                            | AY182241.5057                                                                                                                            |                |      |          |
| KEYWORDS                                                                                                                                                                           |                                                                                                                                          |                |      |          |
| SOURCE                                                                                                                                                                             | Malus x domestica (cultivated apple)                                                                                                     |                |      |          |
| Locus name                                                                                                                                                                         |                                                                                                                                          | Molecule type  |      | Division |
| Phylum: Eukaryota; Viridiplantae; Embryophyta; Tracheophyta; Spermatophyta; Magnoliophyta; eudicotyledons; core eudicots; rosids; eurosids I; Rosales; Rosaceae; Maloideae; Malus. |                                                                                                                                          |                |      |          |
| REFERENCE                                                                                                                                                                          | 1 (bases 1 to 1931)                                                                                                                      |                |      |          |
| AUTHORS                                                                                                                                                                            | Pechous,S.W. and Whitaker,B.D.                                                                                                           |                |      |          |
| TITLE                                                                                                                                                                              | Cloning and functional expression of an (E,E)-alpha-farnesene synthase cDNA from peel tissue of apple fruit                              |                |      |          |
| JOURNAL                                                                                                                                                                            | Planta 219, 84-94 (2004)                                                                                                                 |                |      |          |
| REFERENCE                                                                                                                                                                          | 2 (bases 1 to 1931)                                                                                                                      |                |      |          |
| AUTHORS                                                                                                                                                                            | Pechous,S.W. and Whitaker,B.D.                                                                                                           |                |      |          |
| TITLE                                                                                                                                                                              | Direct Submission                                                                                                                        |                |      |          |
| JOURNAL                                                                                                                                                                            | Submitted (18-NOV-2002) PSI-Produce Quality and Safety Lab, USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD 20705, USA |                |      |          |
| REFERENCE                                                                                                                                                                          | 3 (bases 1 to 1931)                                                                                                                      |                |      |          |
| AUTHORS                                                                                                                                                                            | Pechous,S.W. and Whitaker,B.D.                                                                                                           |                |      |          |
| TITLE                                                                                                                                                                              | Direct Submission                                                                                                                        |                |      |          |
| JOURNAL                                                                                                                                                                            | Submitted (25-JUN-2003) PSI-Produce Quality and Safety Lab, USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD 20705, USA |                |      |          |
| REMARK                                                                                                                                                                             | Sequence update by submitter                                                                                                             |                |      |          |
| COMMENT                                                                                                                                                                            | On Jun 26, 2003 this sequence version replaced gi:27804758.                                                                              |                |      |          |

# Header: Database Identifiers



LOCUS AY182241 1931 bp mRNA linear PLN 04-MAY-2004  
DEFINITION Malus x domestica (E,E)-alpha-farnesene synthase (AFS1) mRNA,  
complete cds.  
ACCESSION AY182241

## Accession

- Stable
- Reportable
- Universal

**ACCESSION AY182241**

**VERSION AY182241.2 GI:32265057**

## Version

Tracks changes in sequence

## GI number

NCBI internal use

Spermatophytes; Magnoliophyta; eudicotyledons; core eudicots;  
Rosales; Rosaceae; Maloideae; Malus  
(1931)  
Whitaker, B.D.  
Cloning and functional expression of an (E,E)-  
synthase cDNA from peel tissue of apple fruit  
JOURNAL Planta 219, 84-94 (2004)  
REFERENCE 2 (bases 1 to 1931)  
AUTHORS Pechous, S.W. and Whitaker, B.D.  
TITLE Direct Submission  
JOURNAL Submitted (18-NOV-2002) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REFERENCE 3 (bases 1 to 1931)  
AUTHORS Pechous, S.W. and Whitaker, B.D.  
TITLE Direct Submission  
JOURNAL Submitted (25-JUN-2003) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REMARK Sequence update by submitter  
COMMENT On Jun 26, 2003 this sequence version replaced gi:27804758.

# Header: Organism



LOCUS AY182241 1931 bp mRNA linear PLN 04-MAY-2004  
DEFINITION Malus x domestica (E,E)-alpha-farnesene synthase (AFS1) mRNA,  
complete cds.  
ACCESSION AY182241  
VERSION AY182241.2 GI:32265057  
KEYWORDS .

SOURCE Malus x domestica (cultivated apple)  
ORGANISM Malus x domestica  
Eukaryota; Viridiplantae; Streptophyta;  
Embryophyta;  
Tracheophyta; Spermatophyta; Magnoliophyta;  
eudicotyledons;  
core eudicots; rosids; eurosids; Rosales;  
Rosaceae;  
Maloideae; Malus.

NCBI-controlled taxonomy

JOURNAL Submitted (18-NOV-2002) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REFERENCE 3 (bases 1 to 1931)  
AUTHORS Pechous,S.W. and Whitaker,B.D.  
TITLE Direct Submission  
JOURNAL Submitted (25-JUN-2003) PSI-Produce Quality and Safety Lab,  
USDA-ARS, 10300 Baltimore Ave. Bldg. 002, Rm. 205, Beltsville, MD  
20705, USA  
REMARK Sequence update by submitter  
COMMENT On Jun 26, 2003 this sequence version replaced gi:27804758.



# The Feature Table



| FEATURES |             | Location/Qualifiers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| source   |             | 1..1931<br>/organism="Malus x domestica"<br>/mol_type="mRNA"<br>/cultivar="'Law Rome'"<br>/db_xref="taxon:3750"<br>/tissue_type="peel"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| gene     |             | 1..1931<br>/gene="AFS1"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| CDS      | start (atg) | 54..1784                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|          |             | /gene="AFS1"<br>/note="terpene synthase"<br>/codon_start=1<br>/product="(E,E)-alpha-farnesene synthase"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|          |             | /protein_id="AA022848.2"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|          |             | /db_xref="GI:32265058"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|          |             | /translation="MEFRVHLQADNEQKIFQNQMKPEPEASYLINQRRSANYKPNIWK<br>NDFLDQSLISKYDGDEYRKLSEKLIIEVKIYISAETMDLVAKLELIDSVRKLGLANLF<br>EKEIKEALDSIAAIESDNLGTRDDLYGTALHFKILRQHGYKVSQDIFGRFMDEKGTLE<br>NHHFAHLKGMLELFEASNLGFEGEDILDEAKASLTALRDSGHICYPDSNLSRDVVHS<br>LELPSHRRVQWFDVKWQINAYEKDICRVNATLLELAKLNFNVVQAQLQKNLREASRWW<br>ANLGIADNLKFARDRLVECFACAVGVAFEPEHSSFRICLTQVINLVLIIDDVYDIYGS<br>EEELKHFTNAVDRWDSRETEQLPECKMKCFQVLYNTTCEIAREIEEENGWNQVLPQLT<br>KVWADFCKALLVEAEWYNKSHIPTLEEYLRNGCISSSVSVLLVHSFFSITHEGTKEMA<br>DFLHKNEDLLYNISLIVRLNNDLGTSAAEQERGDS PSSIVCYMREVNASEETARKNIK<br>GMIDNAWKVNGKCF TTNQVPFLS SFMNNATNMARVAHS L YKDG DGF GDQEKGP RTHI<br>LSLLFQPLVN" |

## GenPept Identifiers



# Common identifiers



- NCBI integrates sequence data from many sources that name sequence records in different ways and where the individual names have different meanings
- Common identifiers for bioinformatics data
  - Locus ID
  - Accession Numbers
  - GenInfo ID
  - PubMed ID
  - ...
- It vary from database to database





# GenInfo (GI) IDs



- Identifier for a particular sequence only
- The GI is assigned in addition to the accession number provided by the source database
- It is simply an integer number, sometimes referred to as a **GI number**.
  - Each entry gets a unique GI number
  - GI: 22477487
  - It provides a single identifier across sequences from many sources
- Not subject to versioning
  - The entry remains the same over all time
  - Different versions of the same sequence are managed by accession numbers
- After the submitter updates a record, even if the sequence differs by a single base pair, it would be given a new GI number





# Appendix B: Data Formats





# FASTA format



- It contains a definition line followed by sequence data
  - No annotation, just sequence
  - For exchange of information between genetic sequence databases
- The definition line starts with a **right angle bracket** (>) and is usually followed by the sequence identifiers in a parsable form, as in this example:
  - >SEQUENCE\_1; comment line 1 (optional)
  - >gi|2352912|gb|AF012433.1|HSDDT2
- The remainder of the definition line is usually a title for the sequence, and can be generated by software from features and other information





# FASTA format



- Many different sequence databases use standardized headers, which helps when automatically extracting information from the header
- Common FASTA header format is given below:
  - GenBank: *gi|gi-number|gb|accession|locus*
  - EMBL Data Library: *gi|gi-number|emb|accession|locus*
  - DDBJ, DNA Database of Japan: *gi|gi-number|dbj|accession|locus*
- Example:

```
>gi|1040960|gb|U35641.1|MMU35641 Mus musculus Brcal mRNA, complete cds
GGCACGAGGATCCAGCACCTCTCTTGGGGCTTCTCCGTCCTCGGCGCTTGGAAGTACGGATCTTTTTTCT
CGGAGAAAAGTTCACTGGAAGTGAAGAAATGGATTTATCTGCCGTCAAATTCAAGAAGTACAAAATGT
CCTTCATGCTATGCAGAAAATCTTAGAGTGTCCGATCTGTTTGGAAGTATCAAAGAACCTGTTTCCACA
AAGTGTGACCACATATTTTGCAAATTTTGTATGCTGAACTTCTTAACCAGAAGAAAGGGCCTTCACAAT
GTCCTTTGTGTAAGAATGAGATAACCAAAAGGAGCCTACAGGGAAGCACAAGGTTTAGTCAGCTTGCTGA
AGAGCTGCTGAGAATAATGGCTGCTTTTGAGCTTGACACGGGAATGCAGCTTACAAATGGTTTTAGTTTT
TCAAAAAAGAGAAATAATTCTTGTGAGCGTTTGAATGAGGAGGCGTCGATCATCCAGAGCGTGGGCTACC
GGAACCGTGTGAGAAGGCTTCCCCAGGTCGAACCTGGAAATGCCACCTTGAAGGACAGCCTAGGTGTCCA
GCTGTCTAACCTTGAATCGTGAGATCAGTGAAGAAAAACAGGCAGACCCAACCTCGAAAGAAATCTGTC
TACATTGAACTAGACTCTGATTCTTCTGAAGAGACAGTAAGCCAGGTGATTGCAGTGTGAGAGACC
...
```





# SWISS-PROT format



- SWISS-PROT database contains reliable protein sequences with annotations (e.g. description of the function of a protein, its domain structure, etc.)
- Each sequence entry is composed of lines
- Each line begins with a 2-character line code (indicating the type of data in the line)

| Line code | Content             | Occurrence in an entry |
|-----------|---------------------|------------------------|
| ID        | Identification      | Once; starts the entry |
| AC        | Accession number(s) | Once or more           |
| DT        | Date                | Three times            |
| DE        | Description         | Once or more           |
| GN        | Gene name(s)        | Optional               |

...

```
ID  BRC1_MOUSE      STANDARD;          PRT;   1812 AA.
AC  P48754; Q60957; Q60983;
DT  01-FEB-1996 (Rel. 33, Created)
DT  01-NOV-1997 (Rel. 35, Last sequence update)
DT  16-OCT-2001 (Rel. 40, Last annotation update)
DE  Breast cancer type 1 susceptibility protein homolog.
GN  BRCA1.
OS  Mus musculus (Mouse).
OC  Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
OC  Mammalia; Eutheria; Rodentia; Sciurognathi; Muridae; Murinae; Mus.
OX  NCBI_TaxID=10090;
RN  [1]
RP  SEQUENCE FROM N.A.
RC  STRAIN=C57BL/6; TISSUE=Embryo;
RX  MEDLINE=96177659; PubMed=8634697;
RA  Abel K.J., Xy J., Yin G.Y., Lyons R.H., Meisler M.H., Weber B.L.;
RT  "Mouse Brcal: localization sequence analysis and identification of
RT  evolutionarily conserved domains.";
RL  Hum. Mol. Genet. 4:2265-2273(1995).
```

...





- XML (**Extensible Markup Language**) is a *de facto* standard for information representation and exchange on the Web
  - Widespread acceptance and ability to handle semi-structured data
- Schema is usually defined in **document type definition** (DTD)
  - Defines valid tags, valid value types
  - Used for validation of format and data
- XSLT (**Extensible Stylesheet Language Transformations**): Defines how to translate documents to other formats

- Example:

```
<?xml version="1.0"?>
<!DOCTYPE GBSeq PUBLIC "-//NCBI//NCBI GBSeq/EN" "http://www.ncbi.nlm.nih.gov/dtd/NCBI_GBSeq.dtd">
<GBSet>
<GBSeq>
  <GBSeq_locus>MMU35641</GBSeq_locus>
  <GBSeq_length>5538</GBSeq_length>
  <GBSeq_strandedness value="not-set">0</GBSeq_strandedness>
  <GBSeq_moltype value="mrna">5</GBSeq_moltype>
  <GBSeq_topology value="linear">1</GBSeq_topology>
  <GBSeq_division>ROD</GBSeq_division>
  <GBSeq_update-date>18-OCT-1996</GBSeq_update-date>
  <GBSeq_create-date>25-OCT-1995</GBSeq_create-date>
  <GBSeq_definition>Mus musculus Brcal mRNA, complete cds</GBSeq_definition>
  <GBSeq_primary-accession>U35641</GBSeq_primary-accession>
  <GBSeq_accession-version>U35641.1</GBSeq_accession-version>
```





# BED format



- UCSC Table Browser
  - Complementary to the UCSC Genome Browser
- **BED (Browser Extensible Data)** format
  - A flexible way to define data lines displayed in an annotation track
  - Tabular output describing the number of elements, the chromosome, start, and stop positions
- Example:

```
browser position chr7:127471196-127495720
browser hide all
track name="ItemRGBDemo" description="Item RGB demonstration" visibility=2 itemRgb="On"
chr7    127471196    127472363    Pos1    0    +    127471196    127472363    255,0,0
chr7    127472363    127473530    Pos2    0    +    127472363    127473530    255,0,0
chr7    127473530    127474697    Pos3    0    +    127473530    127474697    255,0,0
chr7    127474697    127475864    Pos4    0    +    127474697    127475864    255,0,0
chr7    127475864    127477031    Neg1    0    -    127475864    127477031    0,0,255
chr7    127477031    127478198    Neg2    0    -    127477031    127478198    0,0,255
chr7    127478198    127479365    Neg3    0    -    127478198    127479365    0,0,255
chr7    127479365    127480532    Pos5    0    +    127479365    127480532    255,0,0
chr7    127480532    127481699    Neg4    0    -    127480532    127481699    0,0,255
```

